Accuracy-Aware ISCC Optimization for Aerial Edge Inference in SI-TNTNs

Thien Hieu Hoang, Tri Nhu Do, and Georges Kaddoum

***Abstract*—In this study, we investigate integrated sensing, communication, and computation (ISCC) for uplink–downlink coexistence edge inference over stochastic integrated terrestrial and non-terrestrial networks (SI-TNTNs). Ground user equipments sense their environment, compress the raw observation into a feature payload, and partially offload the resulting inference task to hovering unmanned aerial vehicles hosting inference servers in 3D space, while jammers impair both links. Node locations follow homogeneous Poisson point processes. Using matched-filter successive interference cancellation on the uplink and zero-forcing beamforming on the downlink, we formulate a mixed-integer nonlinear program trading off delay, energy, downlink rate, and inference accuracy. Both sensing-induced variables admit closed forms (the three-point delay–energy offloading split and the utility-optimal feature-retention ratio). Accordingly, the ISCC extension leaves the search complexity of the whale/particle-swarm solver unchanged.**

***Index Terms*—ISCC, UAV edge inference, SI-TNTN, partial offloading, inference accuracy, resource allocation, whale optimization.**

I. Introduction

Sixth-generation (6G) networks envisage integrated sensing, communication, and computation (ISCC) as a unified service where the primary figure of merit is inference accuracy rather than raw data rate. Stochastic integrated terrestrial and non-terrestrial networks (SI-TNTNs) [1] naturally host such services via unmanned aerial vehicles (UAVs) deploying edge servers in 3D space, necessitating performance guarantees in distribution over random deployments.

In edge inference, a ground device senses, extracts features, and offloads a fraction to a UAV server. Feature compression directly trades inference accuracy against transmission airtime and computing cycles under a shared resource budget. This coupling is compounded by uplink (UL) and downlink (DL) coexistence: half-duplex UAVs incur cross-link interference, successive interference cancellation (SIC) leaves residual intra-cell interference and decoding margins, zero-forcing beamforming (ZFBF) serves DL UEs, and ground jammers corrupt both directions.

The resulting mixed-integer nonlinear program (MINLP) couples association, power, computing resources, and sensing variables, and is NP-hard [2], motivating swarm heuristics [3], [4]. Prior ISCC studies focused on terrestrial topologies with deterministic geometry and benign spectra [5]–[7]. Conversely, non-terrestrial computing designs [8], [9] considered unidirectional links without jamming or stochastic aerial geometry. None have jointly tackled stochastic aerial geometry, full UL–DL coexistence under jamming, and task-oriented accuracy.

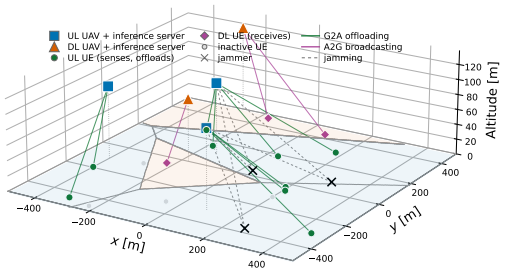


Fig. 1. One HPPP realisation of the SI-TNTN drawn by the simulator: UAVs hover at their sampled altitudes over ground UEs and jammers, with G2A offloading, A2G broadcasting, and jamming in both directions.

Accordingly, in this letter, we recast stochastic UL–DL edge inference in SI-TNTNs as an ISCC problem. Our contributions are: (i) a 3D HPPP aerial edge-inference model with jamming-aware MF-SIC and ZFBF; (ii) an MINLP jointly optimizing delay, energy, DL rate, and inference accuracy; (iii) exact closed-form solutions for the offloading split, feature retention, and compute allocation, incurring zero additional search complexity; and (iv) a hybrid metaheuristic solver benchmarked against exhaustive search and empirical MNIST split-inference.

II. System and Signal Models

Let us consider a square area with M hovering UAVs (each with L antennas and an onboard inference server), ground UEs, and ground jammers (see Fig. 1). UAVs are half-duplex, $M = M^{\mathtt{ul}}+M^{\mathtt{dl}}$. Bandwidth B is reused over K sub-channels of width $B_k = B/K$. Horizontal locations of UAVs, UEs, and jammers are independent 2D HPPPs Φ_0, Φ_{UE}, Φ_{Q} of intensities λ_0, λ_{UE}, λ_{Q}, respectively. Ground nodes sit on $z = z_{\min}$; UAV altitudes are independently and uniformly drawn from $[z_{\mathrm{uav}}^{\min}, z_{\mathrm{uav}}^{\max}]$. UAV m, UE n, and jammer q thus sit at $\vec{\ell}_m, \vec{\ell}_n, \vec{\ell}_q \in \mathbb{R}^3$, respectively. Nearest-UAV association assigns each UE n to UAV $m_n = \arg\min_m \|\vec{\ell}_m - \vec{\ell}_n\|_2$, inducing Voronoi-like cells; the binary $\vec{A}^{\mathtt{ul}}$ then assigns a sub-channel, or local execution, inside the serving cell.

Each slot splits into an orthogonal sensing sub-slot, where UL UEs acquire and compress their observations, and a communication sub-slot carrying UL offloading and DL broadcasting. Accordingly, sensing contends for time and energy, but not spectrum, leaving Sections II-B–II-C unaffected by the sensing waveform.

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UL/DL UAV roles are an independent Bernoulli split with $\lambda_0^{\mathtt{ul}} = \lambda_0^{\mathtt{dl}}$.

For $\psi \in \{\mathtt{G2A}, \mathtt{A2G}\}$, $\vec{h}_{\mathrm{AB}} = \sqrt{\beta_{\mathrm{AB}}^{\psi}}\,\vec{g}_{\mathrm{AB}} \in \mathbb{C}^{L\times 1}$, where $\beta_{\mathrm{AB}}^{\psi} = 10^{-\mathrm{PL}_{\mathrm{AB}}^{\psi}/10}\,\zeta_{\mathrm{AB}}$ is the linear large-scale gain, PL^{ψ} is path loss in dB, ζ_{AB} is inverse-Gamma shadowing, and, finally, $\vec{g}_{\mathrm{AB}}$ is Nakagami-m fading with i.i.d. uniform phases. G2A path loss is $\mathrm{PL}^{\mathtt{G2A}} = 10\alpha\log_{10}(d/d_0) + \mathrm{PL}_0$; A2G is $\mathrm{PL}^{\mathtt{A2G}} = \mathrm{FSPL}(d) + \eta_0$ with $\mathrm{FSPL}(d) = 20\log_{10}(d) + 20\log_{10}(f_c) - 27.55$ for distance d in metres and carrier f_c in MHz, and excess loss η_0. Let us denote $\vec{h}_{n,k,m}$ the UE–UAV channel on sub-channel k.

A. Sensing, Feature Extraction, and Accuracy

Each UL UE $n \in \mathcal{N}^{\mathtt{ul}}$ illuminates its field of view for $\mathsf{T}_n^{\mathtt{sen}}$ at power $p_n^{\mathtt{sen}}$, at cost $\mathsf{E}_n^{\mathtt{sen}} = p_n^{\mathtt{sen}}\mathsf{T}_n^{\mathtt{sen}}$, and collects a raw observation of $D_n^{\mathtt{raw}}$ bits at sensing SNR $\gamma_n^{\mathtt{sen}}$. The sensing configuration $(\mathsf{T}_n^{\mathtt{sen}}, p_n^{\mathtt{sen}})$ is part of the task descriptor; joint sensing-power control is left to future work. The UE then extracts features at ϖ_n cycles/bit, costing $\mathsf{T}_n^{\mathtt{ext}} = \varpi_n D_n^{\mathtt{raw}}/F_n^{\mathtt{loc}}$ and $\mathsf{E}_n^{\mathtt{ext}} = \kappa_n\varpi_n D_n^{\mathtt{raw}}(F_n^{\mathtt{loc}})^2$ with switched-capacitance coefficient κ_n, and retains a fraction $\chi_n \in (0,1]$ of the extracted features. The surviving inference task is thus computed as follows:

$$I_n(\chi_n) = \{\chi_n D_n^{\mathtt{raw}},\ \chi_n C_n^{\mathtt{raw}},\ \gamma_n^{\mathtt{sen}}\}, \tag{1}$$

i.e. retention scales payload and workload alike. Following the discriminant-gain view of task-oriented compression [5], where classification accuracy grows with, and saturates in, the discriminant information carried by the retained features, we adopt the following saturating surrogate (validated empirically on a BottleNet Split-CNN in Section IV):

$$\Lambda_n(\chi_n) = 1 - e^{-\vartheta_n\chi_n\Gamma_n}, \qquad \Gamma_n \triangleq \log_2(1+\gamma_n^{\mathtt{sen}}), \tag{2}$$

with task sensitivity $\vartheta_n > 0$, so that accuracy is increasing and concave in the retained feature information $\chi_n\Gamma_n$ and bounded by one. Therefore, χ_n is the single ISCC knob: raising it improves accuracy, but inflates airtime, transmit energy, and server cycles simultaneously. Imposing $\Lambda_n \geq \Lambda^{\mathtt{th}}$ gives the following retention floor:

$$\chi_n^{\mathtt{min}} = -\ln(1-\Lambda^{\mathtt{th}})/(\vartheta_n\Gamma_n), \tag{3}$$

and UE n is admissible only if $\chi_n^{\mathtt{min}} \leq 1$, i.e. only if its sensing quality satisfies $\vartheta_n\Gamma_n \geq -\ln(1-\Lambda^{\mathtt{th}})$. When $\chi_n^{\mathtt{min}} > 1$, the accuracy constraint is infeasible at any $\chi_n \leq 1$; such UEs leave the constraint set, execute locally, and are assigned zero utility.

B. UL Partial Offloading and MF-SIC

UE n splits its task: a fraction $\rho_n \in [0,1]$ is offloaded and $1-\rho_n$ is executed locally at $F_n^{\mathtt{loc}}$. It has association $a_{n,k}^{\mathtt{ul}} \in \{0,1\}$ with $\sum_k a_{n,k}^{\mathtt{ul}} \leq 1$, and power $p_n \in [0, p_n^{\mathtt{max}}]$. The all-local, full-retention reference costs are computed as follows:

$$\begin{aligned}\mathsf{T}_n^{\mathtt{ref}} &= \mathsf{T}_n^{\mathtt{sen}} + \mathsf{T}_n^{\mathtt{ext}} + C_n^{\mathtt{raw}}/F_n^{\mathtt{loc}},\\ \mathsf{E}_n^{\mathtt{ref}} &= \mathsf{E}_n^{\mathtt{sen}} + \mathsf{E}_n^{\mathtt{ext}} + \kappa_n C_n^{\mathtt{raw}}(F_n^{\mathtt{loc}})^2.\end{aligned} \tag{4}$$

For UE i, sub-channel k, UAV m, let us denote $\mathsf{g}_i \triangleq \|\vec{h}_{i,k,m}\|^2$, $c_{n,i} \triangleq |\vec{h}_{n,k,m}^{\mathsf{H}}\vec{h}_{i,k,m}|^2$, $\mathcal{N}_m^{\mathtt{ul}} = \{n : m_n = m\}$, $\mathcal{N}_k^{\mathtt{sub}} = \{i : a_{i,k}^{\mathtt{ul}} = 1\}$, and weaker intra-cell co-channel UEs $\mathcal{W}_{n,k,m} = \{i \in (\mathcal{N}_m^{\mathtt{ul}} \cap \mathcal{N}_k^{\mathtt{sub}}) \setminus \{n\} : \mathsf{g}_i \leq \mathsf{g}_n\}$. The received vector at UL UAV m on sub-channel k is computed as follows:

$$\begin{aligned}\vec{y}_{k,m} = &\textstyle\sum_{i\in\mathcal{N}_k^{\mathtt{sub}}} \vec{h}_{i,k,m}\sqrt{p_i}s_i + \sum_{m'\in\mathcal{M}_k^{\mathtt{sub}}} \vec{G}_{m',k,m}\vec{P}_{m',k}^{\mathtt{UAV}}\\ &+ \textstyle\sum_{q\in\mathcal{Q}} a_{q,k}\vec{h}_{q,k,m}\sqrt{p_q}s_q + \vec{z}_{k,m},\end{aligned} \tag{5}$$

with $\mathcal{M}_k^{\mathtt{sub}}$ being the set of DL UAVs using sub-channel k with transmit matrices $\vec{P}_{m',k}^{\mathtt{UAV}}$, the UAV-to-UAV channel $\vec{G}_{m',k,m} \in \mathbb{C}^{L\times L}$, jammer activity $a_{q,k} \sim \mathrm{Bernoulli}(q_k)$ (q_k is a probability, distinct from the DL powers $q_{m',n'}$), and $\vec{z}_{k,m} \sim \mathcal{CN}(\vec{0}, N_0\vec{I}_L)$. MF-SIC uses $\vec{v}_{m,n} = \vec{h}_{n,k,m}$ and cancels stronger users first. The Signal-to-Interference-plus-Jamming-and-Noise Ratio (SIJNR) is computed as follows:

$$\gamma_{n,k,m} = \frac{\mathsf{g}_n^2\,p_n}{\mathcal{I}_{n,k,m}^{\mathtt{w}} + \mathcal{I}_{n,k,m}^{\mathtt{o}} + \mathcal{J}_{n,k,m} + \mathsf{g}_n(\Xi_{k,m}/L + B_k N_0)}, \tag{6}$$

where $\mathcal{I}_{n,k,m}^{\mathtt{w}} = \sum_{i\in\mathcal{W}_{n,k,m}} c_{n,i}p_i$ is residual intra-cell interference after SIC, $\mathcal{I}_{n,k,m}^{\mathtt{o}} = \sum_{i\in\mathcal{N}_k^{\mathtt{sub}}\setminus(\mathcal{N}_m^{\mathtt{ul}}\cup\{n\})} c_{n,i}p_i$ is inter-cell UL interference, $\Xi_{k,m} = \sum_{m'\in\mathcal{M}_k^{\mathtt{sub}}} \|\vec{G}_{m',k,m}\vec{P}_{m',k}^{\mathtt{UAV}}\|^2$, and $\mathcal{J}_{n,k,m} = \sum_{q\in\mathcal{Q}} a_{q,k}c_{n,q}p_q$ is jamming, $\mathcal{Q}$ being the jammer set. Since the MF combiner is unnormalized, the projected noise is $\mathsf{g}_n B_k N_0$ and the isotropic DL leakage is $\mathsf{g}_n\Xi_{k,m}/L$; accordingly, both scale with g_n as the numerator does. Reliable SIC requires a decoding margin $\epsilon_m^{\mathtt{th}} \geq 1$,

$$p_n\mathsf{g}_n \big/ \textstyle\sum_{i\in\mathcal{W}_{n,k,m}} p_i\mathsf{g}_i \geq \epsilon_m^{\mathtt{th}}. \tag{7}$$

Users are decoded in decreasing channel order g_n; margin constraint (7) enforces sufficient SINR over uncancelled co-channel users $\mathcal{W}_{n,k,m}$ for reliable SIC (vacuous if $\mathcal{W}_{n,k,m} = \emptyset$). The transmission rate is $R_{n,k,m} = B_k\log_2(1+\gamma_{n,k,m})$, and $a_n^{\mathtt{ul}} = \sum_k a_{n,k}^{\mathtt{ul}}$ with $\rho_n \leq a_n^{\mathtt{ul}}$, so that a UE without a sub-channel computes locally. The local branch and the offload branch run in parallel; thus, with $R_n \triangleq \sum_{m,k} a_{n,k}^{\mathtt{ul}}R_{n,k,m}$ and ignoring result-feedback payload,

$$\mathsf{T}_n = \mathsf{T}_n^{\mathtt{sen}} + \mathsf{T}_n^{\mathtt{ext}} + \chi_n\Delta_n,\ \ \Delta_n = \max\{\Delta_n^{\mathtt{loc}}, \Delta_n^{\mathtt{ofl}}\}, \tag{8}$$

$$\Delta_n^{\mathtt{loc}} = \frac{(1-\rho_n)C_n^{\mathtt{raw}}}{F_n^{\mathtt{loc}}},\ \ \Delta_n^{\mathtt{ofl}} = \rho_n\Big(\frac{D_n^{\mathtt{raw}}}{R_n} + \frac{C_n^{\mathtt{raw}}}{F_{nm(n)}}\Big), \tag{9}$$

$$\mathsf{E}_n = \mathsf{E}_n^{\mathtt{sen}} + \mathsf{E}_n^{\mathtt{ext}} + \chi_n\Theta_n, \tag{10}$$

$$\Theta_n = (1-\rho_n)\kappa_n C_n^{\mathtt{raw}}(F_n^{\mathtt{loc}})^2 + \frac{\rho_n p_n D_n^{\mathtt{raw}}}{\xi_n R_n}, \tag{11}$$

where $\xi_n \in (0,1]$ is the power-amplifier efficiency. $\Delta_n^{\mathtt{loc}}$ and $\Delta_n^{\mathtt{ofl}}$ denote the local and offload computation delays, respectively. Θ_n is UE-side energy only (local compute and UL transmit); UAV server energy is omitted. Setting $\chi_n = 1$ and $\rho_n = a_n^{\mathtt{ul}}$ recovers the conventional atomic-task model. Importantly, χ_n factors out of both branches in Eq. (9), which decouples retention from the split; this is exploited in Section III.

C. DL Broadcasting and ZFBF

DL UAV m' uses one sub-channel, $\sum_k a_{m',k}^{\mathtt{dl}} = 1$, and ZFBF precoder $\vec{W}_{m',k} = \vec{H}_{m',k}^{\mathtt{dl}}[(\vec{H}_{m',k}^{\mathtt{dl}})^{\mathsf{H}}\vec{H}_{m',k}^{\mathtt{dl}}]^{\dagger}$, the Moore–Penrose inverse (where column $\vec{w}_{m',n'}$ is the beamformer for UE n'), which nulls intra-cell interference when $L \geq |\mathcal{N}_{m'}^{\mathtt{dl}}|$, and is regularized otherwise (ensuring spatial nulls when the transmit antenna count L meets the served UE count,

and robust bounded inversion when overloaded). The SIJNR of UE n' is

$$\gamma^{\mathtt{dl}}_{m',k,n'} = q_{m',n'}/(\mathcal{I}^{\mathtt{inter}}_{n'} + \mathcal{I}^{\mathtt{CCI}}_{n'} + \mathcal{J}^{\mathtt{dl}}_{n'} + \sigma^2_{k,n'}), \quad (12)$$

where $q_{m',n'}$ is the DL transmit power allocated to UE n', $\mathcal{I}^{\mathtt{inter}}_{n'} = \sum_{j\neq m'}\sum_{i\in\mathcal{N}^{\mathtt{dl}}_j} q_{j,i}|\vec{h}^{\mathsf{H}}_{n',k,j}\vec{w}_{j,i}|^2$ is inter-cell DL interference, $\mathcal{I}^{\mathtt{CCI}}_{n'} = \sum_{i\in\mathcal{N}^{\mathtt{sub}}_k} p_i|\ell_{i,k,n'}|^2$ is UL-to-DL co-channel interference with G2G channel $\ell_{i,k,n'}$, $\mathcal{J}^{\mathtt{dl}}_{n'} = \sum_q a_{q,k}|h_{q,k,n'}|^2 p_q$ is jamming, and, finally, $\sigma^2_{k,n'} = B_k N_0$ is thermal noise. Then $R^{\mathtt{dl}}_{n'} = \sum_{m',k} a^{\mathtt{dl}}_{m',k} B_k \log_2(1+\gamma^{\mathtt{dl}}_{m',k,n'})$, with total DL power $\sum_{n'} q_{m',n'}\|\vec{w}_{m',n'}\|^2 \le P^{\mathtt{max}}_{m'}$, $\gamma^{\mathtt{dl}}_{m',k,n'} \ge \gamma_{\mathtt{th}}$.

III. Problem and Solution

The UL utility is the weighted sum of normalized delay, energy, and accuracy gains:

$$\mathbb{U}^{\mathtt{ul}}_n = \beta^{\mathtt{t}}_n \frac{\mathsf{T}^{\mathtt{ref}}_n - \mathsf{T}_n}{\mathsf{T}^{\mathtt{ref}}_n} + \beta^{\mathtt{e}}_n \frac{\mathsf{E}^{\mathtt{ref}}_n - \mathsf{E}_n}{\mathsf{E}^{\mathtt{ref}}_n} + \beta^{\mathtt{a}}_n \frac{\Lambda_n - \Lambda^{\mathtt{th}}}{1-\Lambda^{\mathtt{th}}}, \quad (13)$$

with $\beta^{\mathtt{t}}_n + \beta^{\mathtt{e}}_n + \beta^{\mathtt{a}}_n = 1$, while $\Lambda^{\mathtt{th}} \in (0,1)$ is the minimum required inference accuracy threshold. The DL utility is $\mathbb{U}^{\mathtt{dl}}_{n'} = R^{\mathtt{dl}}_{n'}/R_{\mathtt{const}}$ with $R_{\mathtt{const}}$ a reference rate that puts both utilities on one scale. The snapshot objective is $\mathbb{U} = \sum_{n\in\mathcal{N}^{\mathtt{ul}}} \mathbb{U}^{\mathtt{ul}}_n + \sum_{n'\in\mathcal{N}^{\mathtt{dl}}} \mathbb{U}^{\mathtt{dl}}_{n'}$. The HPPP-averaged utility is then computed as follows:

$$\Upsilon_{\mathtt{sys}} = \mathbb{E}[\mathbb{U}]. \quad (14)$$

Writing $\vec{\Pi} \triangleq (\vec{A}^{\mathtt{ul}}, \vec{A}^{\mathtt{dl}}, \vec{p}, \vec{F}, \vec{\rho}, \vec{\chi})$ for the decision tuple, each snapshot solves the MINLP:

$$\max_{\vec{\Pi}} \mathbb{U} \quad (15a)$$

$$\text{s.t. } a^{\mathtt{ul}}_{n,k}, a^{\mathtt{dl}}_{m',k} \in \{0,1\}, \textstyle\sum_k a^{\mathtt{ul}}_{n,k} \le 1, F_{nm} \ge 0, \quad (15b)$$

$$0 \le p_n \le p^{\mathtt{max}}_n, \textstyle\sum_{n\in\mathcal{N}^{\mathtt{ul}}_m} F_{nm} \le F^{\mathtt{max}}_m, \quad (15c)$$

$$0 \le \rho_n \le a^{\mathtt{ul}}_n, \chi^{\mathtt{min}}_n \le \chi_n \le 1, \quad (15d)$$

$$\textstyle\sum_k a^{\mathtt{dl}}_{m',k} = 1, \sum_{n'} q_{m',n'}\|\vec{w}_{m',n'}\|^2 \le P^{\mathtt{max}}_{m'}, \quad (15e)$$

$$q_{m',n'} \ge 0, \gamma^{\mathtt{dl}}_{m',k,n'} \ge \gamma_{\mathtt{th}}, (7). \quad (15f)$$

Constraint (15d), the only structural addition over the atomic-task formulation, absorbs the accuracy constraint $\Lambda_n \ge \Lambda^{\mathtt{th}}$ through Eq. (3). Problem in Eq. (15) inherits NP-hardness from the atomic-task MINLP [2]. We prune associations by Lemma 2, run penalty-augmented power control at the atomic split $(\rho_n, \chi_n) = (a^{\mathtt{ul}}_n, 1)$, and then block-coordinate $\vec{F}$ with $(\vec{\rho}, \vec{\chi})$.

Proposition 1 (Compute allocation): For fixed $(\vec{A}^{\mathtt{ul}}, \vec{p}, \vec{\rho}, \vec{\chi})$, $\mathbb{U}$ is concave in $\vec{F}$. The KKT point of Eq. (15c) is $F^\star_{nm} = 0$ if $a^{\mathtt{ul}}_n \rho_n = 0$, and otherwise the clipped water-filling rule:

$$F^\star_{nm} = \min\left\{\mu_m^{-1/2}\sqrt{\beta^{\mathtt{t}}_n \rho_n \chi_n C^{\mathtt{raw}}_n/\mathsf{T}^{\mathtt{ref}}_n},\ \bar{F}_{nm}\right\}, \quad (16)$$

where

$$\bar{F}_{nm} = \frac{\rho_n C^{\mathtt{raw}}_n}{\left[(1-\rho_n)C^{\mathtt{raw}}_n/F^{\mathtt{loc}}_n - \rho_n D^{\mathtt{raw}}_n/R_n\right]_+} \quad (17)$$

is the CPU frequency beyond which the offload branch of Eq. (9) falls below the local branch (with $\bar{F}_{nm} = \infty$ when the bracket is nonpositive), and $\mu_m > 0$ is the unique water level yielding $\sum_{n\in\mathcal{N}^{\mathtt{ul}}_m} F^\star_{nm} = F^{\mathtt{max}}_m$, or $\mu_m \to 0$ if $\sum_n \bar{F}_{nm} \le F^{\mathtt{max}}_m$.

Proof: $\Delta^{\mathtt{ofl}}_n$ is convex in F_{nm} and $\Delta^{\mathtt{loc}}_n$ constant, so Δ_n is convex and $\mathbb{U}^{\mathtt{ul}}_n$ concave by Eqs. (8), (13); Θ_n, Λ_n are $\vec{F}$-free. Where $\Delta^{\mathtt{ofl}}_n$ is active, $\partial\mathbb{U}^{\mathtt{ul}}_n/\partial F_{nm} = (\beta^{\mathtt{t}}_n/\mathsf{T}^{\mathtt{ref}}_n)\chi_n\rho_n C^{\mathtt{raw}}_n/F^2_{nm} > 0$, so the budget binds; stationarity of $\sum_n \mathbb{U}^{\mathtt{ul}}_n + \mu_m(F^{\mathtt{max}}_m - \sum_n F_{nm})$ yields $F_{nm} \propto \sqrt{\beta^{\mathtt{t}}_n\rho_n\chi_n C^{\mathtt{raw}}_n/\mathsf{T}^{\mathtt{ref}}_n}$. Beyond $\bar{F}_{nm}$, the local branch is active and the derivative vanishes, giving the clip. Monotonicity of $\sum_n F^\star_{nm}$ in μ_m^{-1} gives uniqueness. ∎

Setting $\rho_n = \chi_n = 1$ makes $\bar{F}_{nm} = \infty$, and without sensing overhead $C^{\mathtt{raw}}_n/\mathsf{T}^{\mathtt{ref}}_n = F^{\mathtt{loc}}_n$, which recovers the conventional proportional allocation.

Proposition 2 (Offloading split): For fixed $(\vec{A}^{\mathtt{ul}}, \vec{p}, \vec{F}, \vec{\chi})$ with $a^{\mathtt{ul}}_n = 1$, $\mathbb{U}^{\mathtt{ul}}_n$ is concave and piecewise linear in ρ_n with a single breakpoint at

$$\rho^\star_n = \frac{C^{\mathtt{raw}}_n/F^{\mathtt{loc}}_n}{C^{\mathtt{raw}}_n/F^{\mathtt{loc}}_n + D^{\mathtt{raw}}_n/R_n + C^{\mathtt{raw}}_n/F_{nm(n)}}, \quad (18)$$

where the two branches of Eq. (9) balance. Optimizer $\rho^\diamond_n$ lies in $\{0, \rho^\star_n, 1\}$, and follows from three utility evaluations.

Proof: $\Delta^{\mathtt{loc}}_n$ and $\Delta^{\mathtt{ofl}}_n$ are affine in ρ_n, respectively decreasing and increasing, so Δ_n is convex piecewise linear and the branches cross once, at Eq. (18). As Θ_n is affine and Λ_n is ρ_n-free, Eq. (13) makes $\mathbb{U}^{\mathtt{ul}}_n$ concave piecewise linear with a single kink, maximized there or at an endpoint. ∎

Proposition 3 (Feature retention): For fixed $(\vec{A}^{\mathtt{ul}}, \vec{p}, \vec{F}, \vec{\rho})$, $\mathbb{U}^{\mathtt{ul}}_n$ is strictly concave in χ_n and

$$\chi^\star_n = \min\left\{1, \max\left\{\chi^{\mathtt{min}}_n, -\frac{\ln\Psi_n}{\vartheta_n\Gamma_n}\right\}\right\},\ \Psi_n = \frac{(1-\Lambda^{\mathtt{th}})\Omega_n}{\beta^{\mathtt{a}}_n\vartheta_n\Gamma_n}, \quad (19)$$

where $\Omega_n = \beta^{\mathtt{t}}_n\Delta_n/\mathsf{T}^{\mathtt{ref}}_n + \beta^{\mathtt{e}}_n\Theta_n/\mathsf{E}^{\mathtt{ref}}_n$ is the marginal delay–energy price of one retained feature unit.

Proof: By Eq. (8) and Eq. (10), T_n and E_n are affine in χ_n, so the first two terms of Eq. (13) contribute $-\Omega_n\chi_n$ up to a constant, while Eq. (2) contributes the strictly concave $\beta^{\mathtt{a}}_n(1 - e^{-\vartheta_n\chi_n\Gamma_n} - \Lambda^{\mathtt{th}})/(1-\Lambda^{\mathtt{th}})$. Setting $\partial\mathbb{U}^{\mathtt{ul}}_n/\partial\chi_n = \beta^{\mathtt{a}}_n\vartheta_n\Gamma_n e^{-\vartheta_n\chi_n\Gamma_n}/(1-\Lambda^{\mathtt{th}}) - \Omega_n = 0$ and projecting onto $[\chi^{\mathtt{min}}_n, 1]$ gives Eq. (19). ∎

Propositions 2–3 are in fact jointly optimal for fixed $(\vec{A}, \vec{p}, \vec{F})$: by Eqs. (8)–(10), $\mathbb{U}^{\mathtt{ul}}_n = \text{const} - \chi_n\Omega_n(\rho_n) + \beta^{\mathtt{a}}_n(\Lambda_n - \Lambda^{\mathtt{th}})/(1-\Lambda^{\mathtt{th}})$; therefore, for every $\chi_n > 0$, the maximizing split minimizes the convex piecewise-linear $\Omega_n(\rho_n)$ over $\{0, \rho^\star_n, 1\}$ independently of χ_n, after which Eq. (19) is exact. Eq. (19) is the ISCC tradeoff in closed form: retention decays logarithmically in the marginal delay–energy price Ω_n and grows with sensing quality Γ_n and accuracy weight $\beta^{\mathtt{a}}_n$. As Ω_n aggregates the radio state through R_n and the server state through F_{nm}, sensing fidelity is set jointly with communication and computation, not in advance. While Props. 1–3 yield exact closed forms for each sub-problem under fixed coordinates, overall problem (15) remains NP-hard, motivating metaheuristic search over outer association $\vec{A}$ and powers $\vec{p}$.

For fixed $(\vec{A}^{\mathtt{ul}}, \vec{F}, \vec{\rho}, \vec{\chi})$, the $\vec{p}$-dependent part of $-\sum_n \mathbb{U}^{\mathtt{ul}}_n$ is

$$\mathbb{W} = \sum_{n: a^{\mathtt{ul}}_n = 1} \frac{\eta_n + \theta_n p_n}{R_{n,k,m}}, \quad (20)$$

with weights $\eta_n = \beta^{\mathtt{t}}_n\chi_n\rho_n D^{\mathtt{raw}}_n/\mathsf{T}^{\mathtt{ref}}_n$ and $\theta_n = \beta^{\mathtt{e}}_n\chi_n\rho_n D^{\mathtt{raw}}_n/(\xi_n\mathsf{E}^{\mathtt{ref}}_n)$. The ISCC variables enter only as the product $\chi_n\rho_n$, so $\mathbb{W}$ retains its atomic-task structure and Lemma 1 applies at any fixed (ρ_n, χ_n). Eq. (20) assumes the offload branch active ($\rho_n \ge \rho^\star_n$); on the local branch,

$\eta_n = 0$ and all statements below still hold. The map $\vec{p} \mapsto \mathbb{W}$ is nonconvex, as $R_{n,k,m}$ couples all co-channel powers through Eq. (6). Dropping $\mathcal{I}^{\mathtt{w}}$, $\mathcal{I}^{\circ}$, Ξ, and $\mathcal{J}$ yields the interference-free SIJNR $\tilde{\gamma}_n = \bar{h}_n p_n$ with $\bar{h}_n = \mathrm{g}_n/(B_k N_0)$ and the separable lower bound

$$\widetilde{\mathbb{W}}(\vec{A}^{\mathtt{ul}}, \vec{p}) = \sum_{n:a_n^{\mathtt{ul}}=1} \frac{\eta_n + \theta_n p_n}{B_k \log_2(1 + \bar{h}_n p_n)} \leq \mathbb{W}(\vec{A}^{\mathtt{ul}}, \vec{p}). \quad (21)$$

Lemma 1: For each offloading UE, $\tilde{w}_n(p) = (\eta_n + \theta_n p)/[B_k \log_2(1 + \bar{h}_n p)]$ is quasiconvex on $(0, p_n^{\mathtt{max}}]$. A minimizer $p_n^{\dagger}$ is $p_n^{\mathtt{max}}$ if $\phi_n(p_n^{\mathtt{max}}) \leq 0$, and otherwise the unique root of

$$\phi_n(p) = \theta_n \log_2(1 + \bar{h}_n p) - \frac{\bar{h}_n}{\ln 2} \frac{\eta_n + \theta_n p}{1 + \bar{h}_n p} = 0, \quad (22)$$

which is found by bisection in $\mathcal{O}(\log(p_n^{\mathtt{max}}/\varepsilon))$ steps.

Proof: The sign of $\tilde{w}_n'(p)$ coincides with $\phi_n(p)$. Moreover, $\phi_n'(p) > 0$ on $(0, \infty)$, so ϕ_n is strictly increasing, making $\tilde{w}_n$ strictly quasiconvex. ∎

Pruning must act before $(\vec{\rho}, \vec{\chi})$ are known. Using the interference-free decoupling of Eq. (21), we replace for each offloading UE R_n by $\bar{R}_n = B_k \log_2(1 + \bar{h}_n p_n^{\mathtt{max}})$, F_{nm} by $F_m^{\mathtt{max}}$, and offload energy by $D_n^{\mathtt{raw}} \ln 2/(\xi_n B_k \bar{h}_n)$ via $\log_2(1 + x) \leq x/\ln 2$. Each replacement decreases Δ_n and Θ_n pointwise, so $\bar{\Omega}_n(\rho_n) \leq \Omega_n(\rho_n)$ for all $(\vec{p}, \vec{F})$. With $\bar{u}_n$ the maximum of $\mathbb{U}_n^{\mathtt{ul}}$ under $\bar{\Omega}_n$ from Propositions 2–3, let

$$\mathbb{F}(\vec{A}) = \sum_n \bar{u}_n(\vec{A}^{\mathtt{ul}}) + \mathbb{U}_{\mathrm{ub}}^{\mathtt{dl}}(\vec{A}^{\mathtt{dl}}), \quad (23)$$

with $\mathbb{U}_{\mathrm{ub}}^{\mathtt{dl}}$ an interference-free full-budget upper bound on $\mathbb{U}^{\mathtt{dl}}$. Write $\mathbb{U}^{\star}$ for the incumbent.

Lemma 2 (Pruning): Fix $\vec{A}$. If $\mathbb{F}(\vec{A}) \leq \mathbb{U}^{\star}$, then no $(\vec{p}, \vec{F}, \vec{\rho}, \vec{\chi})$ can achieve a utility strictly greater than $\mathbb{U}^{\star}$.

Proof: $\mathbb{U}_n^{\mathtt{ul}} \leq \bar{u}_n$ for every $(\vec{p}, \vec{F}, \rho_n, \chi_n)$ since $\Omega_n \geq \bar{\Omega}_n$ enters with a negative sign, and $\mathbb{U}^{\mathtt{dl}} \leq \mathbb{U}_{\mathrm{ub}}^{\mathtt{dl}}$; hence $\mathbb{U} \leq \mathbb{F}$. ∎

The remaining continuous problems use penalty-augmented search. For UL power,

$$\min_{\vec{p}} \ \mathbb{W}(\vec{A}^{\mathtt{ul}}, \vec{p}) + \nu_p \sum_n (p_n - p_n^{\mathtt{max}})_+^2 + \nu_{\mathrm{sic}} \sum_{n,k} g_{n,k}^{\mathrm{sic}}(\vec{p})_+^2, \quad (24)$$

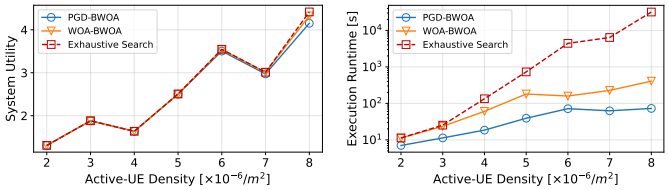
where $g_{n,k}^{\mathrm{sic}}$ is the slack of Eq. (7). For the DL powers $\vec{q} = \{q_{m',n'}\}$, we maximize $\mathbb{U}^{\mathtt{dl}}$ minus squared slacks of Eq. (15e). Both continuous power allocations can be solved via the whale optimization algorithm (WOA) [10], particle swarm optimization (PSO), or projected gradient descent (PGD). To avoid local optima, multi-start PGD is initialized from structured seeds: $\vec{p}^{\dagger}$ (Lemma 1), $\vec{p}^{\mathtt{max}}$, and intermediate power levels in $[\mathbf{0}, \vec{p}^{\mathtt{max}}]$. Binary matching uses BWOA [11]: a whale position $x \in \mathbb{R}$ maps by $\sigma(x) = (1 + e^{-\tau x})^{-1}$ to $a = \mathbf{1}\{\mathrm{rand} < \sigma(x)\}$.

The inner power control costs $\mathcal{O}(I_1 S_1 N^{\mathtt{ul}} K)$ per association that survives Lemma 2; BWOA over $\vec{A}$ costs $\mathcal{O}(I_3 S_3 K(N^{\mathtt{ul}} + M^{\mathtt{dl}}))$ plus the surviving inner solves. The ISCC variables add only $\mathcal{O}(N^{\mathtt{ul}})$ per inner iteration (several block-coordinate rounds of Propositions 1–3), which is dominated by the $\mathcal{O}(I_1 S_1 N^{\mathtt{ul}} K)$ power search, so the overall order remains unchanged from the atomic-task formulation. Exhaustive matching enumerates $(K+1)^{N^{\mathtt{ul}}} K^{M^{\mathtt{dl}}}$ associations.

Algorithm 1: Hybrid WOA/PSO/PGD–BWOA for SI-TNTN UAV edge-inference ISCC.

Input: HPPP snapshot, channels $\vec{h}$, sensed tasks $\{D_n^{\mathtt{raw}}, C_n^{\mathtt{raw}}, \gamma_n^{\mathtt{sen}}\}$, budgets, accuracy floor $\Lambda^{\mathtt{th}}$
Output: $\vec{\Pi} = (\vec{A}^{\mathtt{ul}}, \vec{A}^{\mathtt{dl}}, \vec{p}, \vec{F}, \vec{\rho}, \vec{\chi})$

Force UEs with $\chi_n^{\mathtt{min}} > 1$ in Eq. (3) to local execution.
Initialize BWOA population of binary association matrices.
for *iteration* $t = 1, \ldots, I_3$ *(or until convergence)* **do**
  Prune candidates with $\mathbb{F}(\vec{A}) \leq \mathbb{U}^{\star}$ by Lemma 2.
  Batch evaluate surviving candidates $\vec{A}$ simultaneously:
    Set $(\rho_n, \chi_n) \leftarrow (a_n^{\mathtt{ul}}, 1)$ across the batch.
    Run batched WOA/PSO/PGD on $\vec{p}$ with SIJNR/SIC penalties.
    Alternate Eq. (16) with Propositions 2–3 until $(\vec{F}, \vec{\rho}, \vec{\chi})$ stabilizes.
    Evaluate batch utilities $\mathbb{U}(\vec{\Pi})$ and update $\mathbb{U}^{\star}$.
  Update BWOA positions via encircling/spiral search and retain elite.
return the best $\vec{\Pi}$.



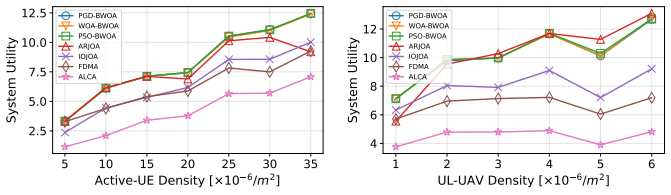
(a) (b)

Fig. 2. Proposed BWOA matching inner PGD and WOA against exhaustive search, $K = 3$: (a) attained utility, (b) runtime, vs. active-UE density.

IV. Numerical Results

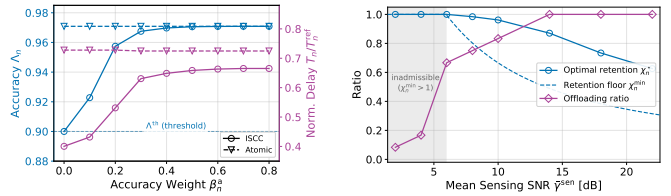
Unless noted otherwise, $K = 5$, $B_k = 1$ MHz, $L = 4$, $N_0 = -174$ dBm/Hz, $f_c = 2.4$ GHz (FSPL uses 2400 MHz), $F_m^{\mathtt{max}} = 8$ GHz, $p_n^{\mathtt{max}} = 24$ dBm, $P_{m'}^{\mathtt{max}} = 46$ dBm, $C_n^{\mathtt{raw}} = 10^9$ cycles, $D_n^{\mathtt{raw}} = 420$ kbit, $F_n^{\mathtt{loc}} \in \{0.5, 0.8, 1.0\}$ GHz, $\kappa_n = 5 \times 10^{-27}$, $\xi_n = 1$, $R_{\mathtt{const}} = 10^8$ bit/s, and $\beta_n^{\mathtt{t}} = \beta_n^{\mathtt{e}} = \beta_n^{\mathtt{a}} = 1/3$. Channels use Nakagami-$m$ ($m = 2$), inverse-Gamma shadowing of shape 3, $\alpha = 2.5$, $\mathrm{PL}_0 = 40$ dB, $d_0 = 1$ m, $\eta_0 = 20$ dB. In 1000×1000 m^2: $\lambda_0^{\mathtt{ul}} = 2, \lambda_0^{\mathtt{dl}} = 1$, $\lambda_{\mathrm{UE}} = 20$, $\lambda_{\mathrm{Q}} = 2$ nodes/km^2, UAV altitude $[80, 120]$ m, $p_q = 20$ dBm, $q_k = 0.8$. The sensing/ISCC parameters are $\gamma_n^{\mathtt{sen}} = 10$ dB (per-UE spread 3 dB), $\mathsf{T}_n^{\mathtt{sen}} = 5$ ms, $p_n^{\mathtt{sen}} = 0.1$ W, $\vartheta_n = 1$, $\Lambda^{\mathtt{th}} = 0.9$, and $\varpi_n = 50$ cycles/bit, which by Eq. (3) yields a retention floor $\chi_n^{\mathtt{min}} \approx 0.67$. We benchmark the hybrids against exhaustive matching and against the atomic-task WOA/PSO/PGD–BWOA framework of [2], with force-all-offload (every UL UE must offload), independent offloading (IOJOA: each UE freezes a coin-flip offload bit), orthogonal access (FDMA: at most one UL UE or DL UAV per sub-channel), and all-local computing (ALCA) as additional baselines. Each metaheuristic uses 30 search agents, and every curve averages 30 HPPP snapshots.

Fig. 2(a) shows both BWOA variants closely tracking exhaustive sub-channel search, with a small gap widening with the growth of the association space. Fig. 2(b) shows the corresponding runtime: exhaustive enumeration grows steeply, whereas both BWOA variants scale near-linearly and maintain substantial time savings with an increase of density. Comparing solver complexity against the atomic baseline, since Props. 1–3 evaluate elementary closed-form expressions requiring $\mathcal{O}(1)$ operations per user, the ISCC updates avoid



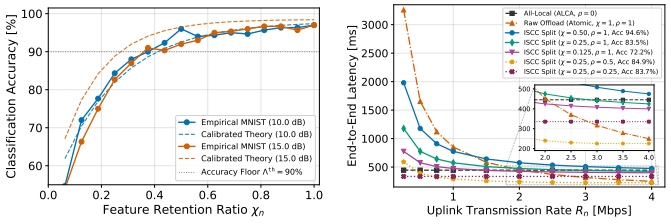
(a) (b)

Fig. 3. (a) Utility versus active-UE density for WOA/PSO/PGD–BWOA (MF-SIC), force-all-offload, IOJOA, FDMA, and ALCA. (b) Utility vs. UL UAV-server density.



(a) (b)

Fig. 4. ISCC effects: (a) accuracy–delay frontier swept over $\beta^{\mathtt{a}}$, vs. the atomic-task baseline; (b) optimal retention $\chi_n^\star$, retention floor $\chi_n^{\min}$, and ratio of offloading UEs vs. mean sensing SNR.



(a) (b)

Fig. 5. Task-oriented edge inference evaluation on MNIST with BottleNet Split-CNN: (a) empirical classification accuracy vs. calibrated theoretical surrogate $\Lambda_n(\chi_n)$ under sensing noise; (b) end-to-end inference latency vs. uplink transmission rate across feature retention ratios χ_n and offload splitting ratios ρ_n.

continuous search and add only basic scalar arithmetic per iteration. Consequently, average runtimes remain on par with the atomic counterpart for 10 UEs (≈ 126 s vs. ≈ 129 s), and are actually faster for 20 UEs (≈ 524 s vs. ≈ 766 s) as feature compression and partial offloading alleviate radio congestion and server bottlenecks.

Fig. 3(a) shows aggregate utility rising with UE density, more terms entering $\mathbb{U}$ even as the co-channel and jamming terms of Eqs. (6)–(12) grow. Fig. 3(b) shows added UAV inference servers changing utility through extra compute and extra G2A/A2G interference.

Fig. 4(a) sweeps $\beta^{\mathtt{a}}$ with $\beta^{\mathtt{t}} = \beta^{\mathtt{e}} = (1-\beta^{\mathtt{a}})/2$. At low $\beta^{\mathtt{a}}$, retention sits on its floor and accuracy equals $\Lambda^{\mathtt{th}}$, achieving the lowest latency; raising $\beta^{\mathtt{a}}$ drives $\chi_n^\star \to 1$, trading latency for accuracy along Eq. (19). While accuracy reaches the atomic-task level at large $\beta^{\mathtt{a}}$, ISCC preserves a lower delay throughout, as partial offloading ($\rho_n \in [0,1]$) enables parallel local-and-aerial execution that the binary atomic model ($\rho_n \in \{0,1\}$) cannot exploit. In Fig. 4(b), at low sensing SNR the floor (3) exceeds one, UEs become inadmissible, and offloading collapses; above it, better sensing lowers both $\chi_n^{\min}$ and $\chi_n^\star$: higher-quality observations are compressed *more*, a coupling an atomic-task model cannot express.

Fig. 5(a) validates the analytical surrogate $\Lambda_n(\chi_n)$ against empirical BottleNet Split-CNN classification on MNIST under sensing noise. The empirical curves closely mirror the characteristic concave, saturating profile formulated in Eq. (2), capturing the diminishing returns of deep feature maps: accuracy ascends steeply at low retention and plateaus once $\chi_n \geq 0.5$. Fig. 5(b) evaluates end-to-end latency over uplink rates R_n. Under constrained bandwidth ($R_n \leq 1.0$ Mbps), ISCC split computing ($\chi = 0.25$) cuts latency by up to $3.5\times$ over raw offloading and $4.2\times$ over local computing, while partial task splitting ($\rho = 0.5$) balances parallel execution to avoid transmission bottlenecks.

V. Conclusion

In this letter, we recast joint UL–DL edge-inference offloading and radio resource management on a SI-TNTN as an ISCC problem combining UAV-mounted servers, planar HPPP geometry with independent altitudes, and jamming-aware MF-SIC/ZFBF SIJNR with sensing-derived tasks, partial offloading, and an inference-accuracy objective. The two sensing-induced variables admit closed forms (a three-point rule for the split and a logarithmic rule for retention). Accordingly, the MINLP is still solved by the WOA/PSO/PGD plus BWOA framework at unchanged complexity. Joint sensing-power control and sensing/communication antenna partitioning at the UAV are natural extensions.

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